

Build a control chart

The exact demo prompt: paste your date-ordered numbers, name the three-standard-deviation rule, and let AI compute the limits and flag the signals.

The prompt

You are a patient statistics tutor. I am Priya, a white belt monitoring our ED improvement. Here are thirty days of daily average door-to-doctor time in minutes, in date order. Please create a control chart. Calculate the center line and the upper and lower control limits using three standard deviations. Plot the points, then tell me whether the process is in statistical control, flagging any points outside the limits or any other out-of-control patterns, like a long run of points on one side of the center line.

What good output looks like

AI returns the chart with a center line at the new average (around thirty four minutes) and the upper and lower control limits calculated from the data itself (around forty six and twenty two), then gives the plain-language verdict. In the in-control base case: the process appears to be in statistical control, all points fall within the limits, no unusual runs or trends, the variation is ordinary common-cause variation. In the alternate case where Day 28 spiked to forty nine: the AI flags it directly, the point on day twenty eight at forty nine minutes is above the upper control limit of forty six, indicating a special cause worth investigating. The chart rings the alarm; the AI cannot know that Day 28 was when two triage nurses were out sick, which is Priya's knowledge to supply.